

Advanced Interpolation Methods

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Introduction

This report aims to summarize the topics covered in MAT 425 Advanced Interpolation Methods. The course began with simple 1 dimensional interpolation methods to form a background understanding of interpolation and its uses. From there the main goal of the course was to understand and replicate the ideas of MRI interpolation presented to us by two research papers. This involved covering and reviewing many different mathematics topics and their applications such as calculus, linear algebra, analysis, complex variables, and number theory. We will connect the ideas of this class to see how we have gone from 1D interpolation to accurate brain tractography imaging.

1 1D Interpolation Methods

Interpolation is defined as an estimation that involves the construction of new data points based on separate known data points.

1.1 Lagrange

The first 1D interpolation method discussed in class was Lagrange Interpolation. This method uses a set of ordered pairs to create a polynomial that can be evaluated at different values to interpolate from a given dataset.

Suppose we are given $n + 1$ distinct data points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

The Lagrange interpolation polynomial $P(x)$ is defined as:

$$P(x) = \sum_{j=0}^n y_j \cdot L_j(x)$$

where $L_j(x)$ is the j -th Lagrange basis polynomial:

$$L_j(x) = \prod_{\substack{0 \leq m \leq n \\ m \neq j}} \frac{x - x_m}{x_j - x_m}$$

1.2 Newton's Divided Difference

The second 1D interpolation method discussed was Newton's Divided Difference. This uses a table method to calculate interpolating polynomials from a given dataset where each divided difference along the diagonal provides the coefficient for the interpolating polynomial.

Given the data points:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

The divided difference table is constructed as follows:

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$\dots$
x_0	y_0			
x_1	y_1	$\frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_2	y_2	$\frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$\frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$	
x_3	y_3	$\frac{f[x_3] - f[x_2]}{x_3 - x_2}$	$\frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$	$\frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

1.3 Cubic Spline

The third 1D interpolation technique we learned was cubic spline interpolation. Cubic spline interpolation uses cubic polynomials between ordered pairs where the first and second derivatives are continuous in order to avoid erratic oscillations and ensure smoothness and continuity.

The conditions for cubic splines are as follows:

1. $S(x)$ is a cubic polynomial, denoted by $S_j(x)$, on the interval $[x_j, x_{j+1}]$ for each $j = 0, \dots, n - 1$.

2. $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1}), j = 0, \dots, n-1$.
3. $S_j(x_{j+1}) = S_{j+1}(x_{j+1}), j = 0, \dots, n-2$.
4. $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1}), j = 0, \dots, n-2$.
5. $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1}), j = 0, \dots, n-2$.
6. One of the following sets of boundary conditions (BCs) is imposed:
 - (a) $S''(x_0) = S''(x_n) = 0$, natural or free BC
 - (b) $S''(x_0)$ and $S'(x_n)$ are equal to some values; clamped BC

2 Paper 1: Angular Resolution Enhancement Technique for DWI

Diffusion-weighted imaging (DWI) is a magnetic resonance imaging technique that is used to study the structure of white matter in the brain. However, methods such as diffusion tensor imaging (DTI) are limited—they only use a few gradient directions, which fails to capture complex fiber structures. To improve this, high-angular-resolution diffusion imaging (HARDI) methods have been developed, but they require long scan times.

This paper proposes a technique to enhance the angular resolution without additional scanning. By analyzing patterns in the raw HARDI signal, the method predicts new diffusion directions and synthesizes the corresponding signals using local signal similarity. These synthesized signals are then used to compute orientation distribution functions (ODFs) with improved accuracy.

2.1 Theory and Method

The core idea is to use known signals in measured diffusion gradient (DG) directions to estimate signals in unmeasured directions. These predictions rely on the idea that, within a local region of space, diffusion behaves similarly along neighboring directions.

For each unmeasured target direction, we then select a set of measured neighboring directions from the DG direction set $\mathcal{D}_N$. From this, triangles such as $\mathcal{N}_{\mathbf{v}_j}^{3,1} \subset \mathcal{D}_N$ are formed to define local geometric neighborhoods. The reliability of a triangle is assessed using correlation coefficients computed across a $3 \times 3 \times 3$ local window. These coefficients evaluate the similarity between the local diffusion patterns in the directions under consideration. If all pairwise correlations within a triangle are positive, it is deemed suitable for interpolation.

A new appropriate direction $\mathbf{u}_c$ is then determined as a weighted combination of the triangle's directions, lying within their convex hull:

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{jk}, \quad \text{where } \sum_{k=1}^K w_k = 1, \quad w_k \geq 0$$

Once the direction is predicted, the signal at $\mathbf{u}_c$ is synthesized by taking a corresponding weighted combination of the signals in the neighboring directions:

$$S(\mathbf{u}_c) = \sum_{k=1}^K w_k S(\mathbf{g}_{jk})$$

To extend the coverage of the signal over a finer angular grid, we then apply RBF interpolation. A Gaussian kernel is used to interpolate the signal at new directions using both measured and synthesized directions:

$$S(\mathbf{v}_j) = \sum_{g \in D_C} c_g \cdot \exp(-\alpha \|\mathbf{v}_j - g\|^2)$$

The interpolation weights $\mathbf{c}_g$ are solved from the linear system:

$$(A + \lambda I)\mathbf{c} = \mathbf{b}$$

2.2 Validation

To evaluate the effectiveness of the proposed interpolation method, the authors compare reconstructed spin distribution functions (SDFs) derived from three datasets:

1. f_{64} : Full measured signal (64 directions)
2. f_{32} : Partial measured signal (32 directions only)
3. f_{cm} : Partial signal + interpolated directions (32 + synthesized)

2.3 Relative L^2 Error

This is the primary quantitative metric:

$$\text{RE}_{32} = \frac{\|\tilde{f}_{64} - \tilde{f}_{32}\|}{\|\tilde{f}_{64}\|}, \quad \text{RE}_{cm} = \frac{\|\tilde{f}_{64} - \tilde{f}_{cm}\|}{\|\tilde{f}_{64}\|}$$

This measures how close the estimated SDFs are to the true distribution.

Crossing Fiber Voxels

The authors also calculate the percentage of crossing fibers that were detected in the brain from each datasets. Interpolated signals (f_{cm}) yield a higher percentage of accurately identified crossings than partial signals alone.

Angular Deviation and Fiber Ratio

The angular deviation between the estimated and true secondary fiber directions is calculated to assess directional fidelity. In Table 3 of the paper, the ratio of second fiber directions F_{d2} is reported for both f_{cm} and f_{32} relative to f_{64} , showing improved structural alignment using interpolated data.

Qualitative Comparison

Figure 4 of the paper displays voxel-wise visualizations of SDFs and their relative errors. These figures support the quantitative findings by highlighting spatial consistency improvements across the various data.

3 Paper 2: Q-Ball Imaging

The main goal of paper 2 is to propose a solution for Q-ball imaging reconstruction of the orientation distribution function (ODF). The ODF is function that is used to find the directions of particles in a material. Simply put, the ODF helps find how likely it is something is pointing in a certain direction. This is important in brain MRI imaging in order to determine the diffusion directions of water molecules. This gives a better insight into the white matter fiber crossings and directions in the brain. It is important to understand these intricate crossings of neuron axons to better understand brain connectivity and the parts of brain that may be affected by certain diseases.

3.1 Theory

The theory section of this paper highlights how this goal is achieved. First the ODF is written as $G[S](\mathbf{u}) = \int_{|w|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) dw$, where G is the diffusion gradient(DG) vector, δ the gradient pulse duration, and $S(\mathbf{u})$ the signal.

The paper then continues to explain how they developed a way to estimate the function on a q-ball that best approximates discrete and sparse measurements using SHs. The definition of a spherical harmonic (SH) is provided to which they reindex and write in a way that they are only looking at the real values of this SH since we do not

care about complex data. Symmetry is ensured by only looking at even SH. Each vector is made orthonormal by taking the inner product over the unit sphere. Each signal is approximated at each of the DG directions as $S(\theta_i, \phi_i) = \sum_{j=1}^R c_j Y_j(\theta_i, \phi_i)$, where R is the number of terms in the modified SH basis Y of order l. Based on their experiments they use $R = \frac{(l+1)(l+2)}{2}$. Letting S be the Nx1 vector representing the input signal for every gradient direction, C being the Rx1 vector of SH coefficients and B as the NxR matrix with the modified SH basis. This can be written as a linear system of equations $S = BC$ where S is all the input signals, B is the discrete SH basis elements, and the c_j coefficients are what they are trying to solve for, where $c_j = \int_{\Omega} S(\theta, \phi) Y_j(\theta, \phi) d\Omega$.

Since this is a tough integral to evaluate, the paper adds local regularization in order to use a high order estimation without over-modeling small perturbations because of noise in the input MRI signal. This is done using the Laplace-Beltrami operator (Δ_b) to parameterize the function. Note that the Laplace-Beltrami operator is an extension of the Laplacian operator to functions defined on surfaces. Thus they define a measure, E, of the deviation from smoothness of a function f on the unit sphere as $E(f) = \int_{\Omega} (\Delta_b f)^2 d\Omega$. Using linear algebra and calculus this can be rewritten. From calculus properties of uniform convergence, Cauchy products, and the linearity of the Laplace operator the result is:

$$E(f) = \int_{\Omega} \Delta_b \left(\sum_p c_p Y_p \right) \Delta_b \left(\sum_q c_q Y_q \right) d\Omega =$$

$$\sum_{j=1}^R c_j^2 l_j^2 (l_j + 1)^2 = C^T L C,$$

where L is the the RxR matrix with entries along the diagonal. This can be expressed in matrix form as $M(C) = (S - BC)^T (S - BC) + \lambda C^T L C$, where λ is the weight on the regularization term. Using properties of linear algebra the generalized expression for the desired SH series coefficient vector then becomes: $C = (B^T B + \lambda L)^{-1} B^T S$. From this the SH coefficient of the signal on the q-ball for any (θ, ϕ) can be recovered.

The authors then prove a corollary to the Funke-Hecke theorem needed to solve the integral. They do work providing a way to write a 3D surface integral as a 1D integral. Then replacing the original ODF to be written using the signal from the SH approximations. So that:

$$G[S](\mathbf{u}) = \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) S(\mathbf{w}) d\mathbf{w} = \sum_{j=1}^R c_j \int_{|\mathbf{w}|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) d\mathbf{w}.$$

This can not directly be evaluated since the Dirac delta function is discontinuous at zero. An approximation using limits of the Dirac delta function must be made in order to evaluate. The paper uses the Gaussian given by $\delta_n(x) = (n/\sqrt{\pi})\exp(-n^2x^2)$ to do this. Using properties of convergence and the funke-hecke theorem, they obtained:

$$\int_{|w|=1} \delta(\mathbf{u}^T \mathbf{w}) Y_j(\mathbf{w}) dw = 2\pi P_l(0) Y_j(u),$$

where $P_l(0)$ is the Legendre Polynomial of degree l evaluated at 0.

Using combinatorics techniques it can be obtained that when l is odd $P_l(0)$ is 0 and when l is even $P_l(0)$ is $(-1)^{\frac{l}{2}} \frac{1*3*5*\dots*(l-1)}{2*4*6*\dots*l}$. Now this can be applied to the ODF:

$$G[S](\mathbf{u}) = \sum_{j=1}^R 2\pi P_{l_j}(0) c_j Y_j(\mathbf{u})$$

Now they are ready to evaluate in matrix form $S = BC$ solving for the c_j coefficients which gives the signal values for the ODF with limited noise and regularized terms giving smooth data allowing for creation of the q-ball imaging for each signal at each direction.

4 MATLAB Algorithms and Images

4.1 Input

- `gdirt` $\in \mathbb{R}^{240 \times 3}$: Diffusion gradient directions on the unit sphere.
- `rho4000` $\in \mathbb{R}^{96 \times 96 \times 40 \times 240}$: 4D diffusion-weighted MRI data.
- `idx` $\subset [1, 240]$: Indices of red (measured) directions; the rest are blue (unmeasured).
- `ii`, `ij`, `nth_sl`: Selected voxel location (e.g., `ii=23`, `ij=12`, `nth_sl=24`).
- λ : Regularization parameter for SH and RBF methods.
- `rbft( $\varphi(r)$ )`: Type of radial basis function (e.g., Gaussian).

4.2 Output

- `estimated_bs`: Interpolated signals at blue directions using RBF.
- `ODFtS`: ODF computed from full signal.
- `ODFvS`: ODF computed from red (measured) directions only.

4.3 Algorithm Steps

1. **Load Data:** Import `gdirt` and `rho4000`.
2. **Direction Partition:** Split the 240 directions into red (measured) and blue (to be estimated).
3. **Signal Extraction:** At voxel (ii, ij) on slice `nth_sl`, extract:
 - `tS`: signal at all 240 directions.
 - `vS`: signal at red directions.
 - `uS`: ground truth signal at blue directions (for validation).
4. **Get Green Neighbors:** Choose measured red directions `vgdirt` and their signals `vS`. Use `get_cm()` to obtain green directions `cmdirt` and signals `cmval`.
5. **Prepare RBF Input:** Combine red, green, and their symmetric directions as interpolation basis. Combine corresponding signal values. Define full input directions including symmetric ones:

```
rg_directions = [vgdirt; -vgdirt; cmdirt; -cmdirt]
```

Corresponding signals:

```
rg_signal = [vS; vS; cmval; cmval]
```

6. **Optimize Epsilon:**

Use `L00CV3Dmin()` to find optimal RBF shape parameter using leave-one-out cross-validation (to find ϵ)

```
ep_opt01 = L00CV3Dmin(rg_directions, rg_signal, rbft)
```

7. **Interpolate Signals**

Apply `RBFInterpolation()` to estimate signal values at blue directions.

```
estimated_bS = RBFInterpolation(rg_directions, rg_signal, ugdirt, rbft, ep_opt01,  $\lambda$ )
```

8. **Compute ODFs**

Use full signal set (measured red+estimated blue) to compute an ODF:

```
ODF = mkODF([vgdirt; ugdirt], [vS; estimated_bS],  $\lambda$ )
```

- `ODFtS`: from full signal.
- `ODFvS`: from red only.

Visualize using `plotQBI`.

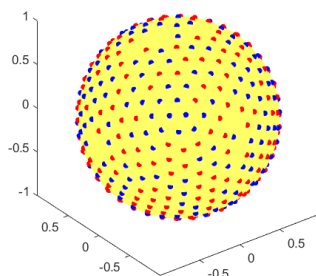


Figure 1: Q-ball

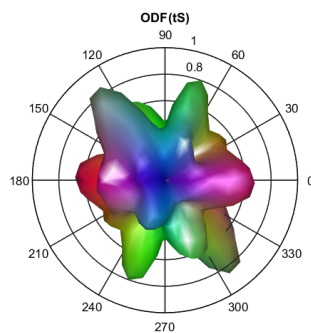


Figure 2: ODF from full signal values

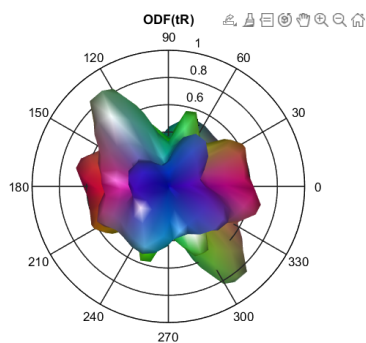


Figure 3: ODF from only red signal values

5 Brain Tractography

Brain tractography is a post-processing technique used in diffusion MRI to reconstruct the probable pathways of white matter fiber tracts in the brain. Using Diffusion Spectrum Imaging (DSI) software, we can model the complex structure of axonal pathways, including areas with fiber crossings. Using the improved ODFs generated from the techniques used in both papers adding more directions and approximating their signal values, we can more accurately estimate local fiber orientations allowing for a tractography to be created. Tractography algorithms then use these ODFs across brain voxels to trace the likely trajectories of neural tracts. While tractography does not provide a direct image of axons, it serves as a valuable tool for visualizing structural connectivity in the brain, supporting diagnosis, research, and surgical planning.

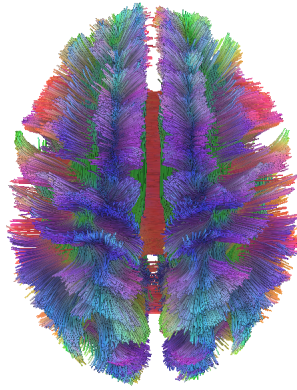


Figure 4: Superior View of Brain Tractography

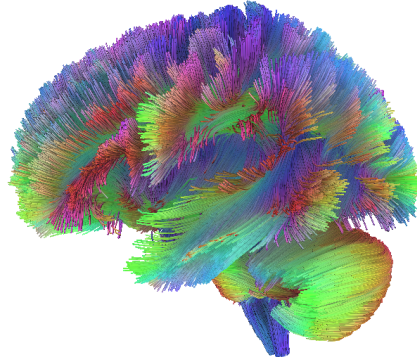


Figure 5: Lateral View of Brain Tractography

6 Reflection and Synthesis

Synthesis

MRI interpolation is necessary due to the time limitations a patient can remain in an MRI machine. In this allotted time not enough data is collected to provide detailed and accurate enough images of the intricate axonal pathways in the white matter of the brain. So we need to use interpolation to build better images with probable axon pathways. More signals are synthesized on q-ball spheres based on the signals we could collect using RBF interpolation. From there the q-balls are stretched in the directions of the signals using SH interpolation to make the ODFs for each voxel. Then the ODFs are combined in brain tractography to create detailed images of probable white matter pathways. This is all extremely helpful and useful to medical professionals when visualizing structural connectivity in the brain, supporting diagnosis, research, and surgical planning.

Reflection

The group faced many challenges regarding our understanding of both papers. Specifically, we struggled with Spherical Harmonic Interpolation and MATLAB Assignments. Through hours spent in your office and late nights, we slowly began to deepen our understanding of the difficult mathematical topics covered in this class. So now, after having gone through this process, we have gained skills that will be integral to our ability to problem-solve throughout our professional careers and beyond.